Architectural Responsibility Configuration (ARCM): A Responsibility-Oriented Theory for Multi-View Architecture Descriptions

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Multi-view architecture descriptions are widely used to manage complexity, yet architecture documentation frequently degrades even when all relevant viewpoints and concerns are covered [12]. Existing coverage-oriented theories fail to explain this phenomenon because they remain silent on where representational authority resides and how relations across views are epistemically maintained. This paper introduces Architectural Responsibility Configuration (ARCM), a responsibility-oriented theory that elevates *representational responsibility*—the assignment of epistemic authority over architectural facts and relations—to a first-class theoretical primitive, following a theory-building approach in software engineering [28].ARCM yields a closed, falsifiable theory that explains documentation decay and stability as predictable outcomes of responsibility configurations under epistemic gravity and evolutionary pressure.

CCS Concepts: • **Software and its engineering → Software architectures**; *Software design engineering*.

Additional Key Words and Phrases: software architecture, architecture description, ISO 42010, responsibility, theory, documentation

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1 Introduction

1.1 Architecture Documentation as an Epistemic System

Software architecture documentation is commonly treated as a representational artifact: a structured collection of models intended to describe a system. In practice, however, architecture documentation functions as an epistemic device. It mediates understanding, supports decision-making, and coordinates evolution across heterogeneous stakeholders over time [2, 7].

Multi-view architecture descriptions play a central role in this mediation. By separating concerns into distinct viewpoints, they enable focused reasoning and reduce local cognitive load [16]. This separation mechanism is both necessary and effective; it explains the enduring success of viewpoint-based standards and frameworks.

However, multi-view descriptions are inherently *divergent*. Each viewpoint emphasizes different abstractions, assumptions, and evaluative criteria. While viewpoints excel at differentiation, they do not, by themselves, provide a theory of integration [18]. The problem of how architectural knowledge converges into a coherent and authoritative understanding is left largely implicit.

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This paper does not challenge the validity of multi-view architecture. Instead, it argues that existing theory under-specifies the epistemic mechanisms by which multi-view representations are stabilized, trusted, and evolved as a system grows in scale and complexity.

1.2 The Persistent Failure of Coverage-Oriented Frameworks

Architecture frameworks typically assess documentation quality in terms of coverage: whether relevant concerns are identified, viewpoints are defined, and corresponding artifacts are produced [12]. Under this assumption, increasing coverage should monotonically improve architectural understanding and utility.

Empirical practice repeatedly contradicts this expectation. Architecture descriptions with extensive viewpoint coverage often exhibit poor comprehensibility, weak authority, and limited usefulness in guiding change [3, 18, 34]. Stakeholders struggle not because information is missing, but because they cannot determine which artifacts are authoritative, how views relate, or where to resolve inconsistencies.

These observations constitute an anomaly for coverage-oriented theory. They do not challenge the correctness of individual viewpoints, nor do they imply insufficient modeling effort. Rather, they indicate a theoretical gap: coverage alone does not explain how architectural knowledge maintains coherence, trust, and stability over time.

Importantly, this claim is not an empirical generalization about specific projects. It is an explanatory failure: existing theory lacks the constructs needed to account for these recurrent breakdowns.

1.3 The Missing Dimension: Responsibility Configuration

We argue that the persistent failure of architecture documentation arises from a missing theoretical dimension: the configuration of representational responsibility.To make this dimension explicit and analytically tractable, this paper introduces Architectural Responsibility Configuration (ARCM), a responsibility-oriented theory that characterizes how epistemic authority over architectural facts and inter-view relations is distributed, anchored, and evolves across architecture artifacts in multi-view descriptions.

The problem is not that viewpoints are insufficiently fine-grained, nor that concerns are poorly identified. Instead, responsibilities for representing architectural knowledge—and critically, for representing the *relations between views*—are rarely made explicit.

In the absence of explicit responsibility configuration, multiple artifacts may implicitly claim authority over the same concern, relational knowledge may be duplicated or omitted, and stakeholders are left without a stable epistemic anchor. These failures occur even when coverage requirements are fully satisfied.

This reframing shifts the core question of architecture documentation from "Which views are needed?" to "How are epistemic responsibilities distributed and anchored across artifacts?", echoing classic observations on structure–communication coupling without reducing the problem to organizational charts [8].

1.4 Contributions

This paper makes the following contributions:

- It introduces Architectural Responsibility Configuration (ARCM), a responsibility-oriented theory that elevates representational responsibility to an irreducible theoretical primitive for reasoning about architecture documentation.
- It identifies recurrent responsibility pathologies and explains them as structural outcomes rather than documentation errors.

- It derives a set of falsifiable propositions governing the evolution and stability of responsibility configurations.
- It characterizes three stable responsibility configurations—Unified Core, Regulatory Overlay, and Federated Configuration—as equilibrium responses to distinct epistemic tensions.
- It provides an evolutionary account of documentation decay and stabilization grounded in epistemic gravity and knowledge metabolism.

2 Background and Motivation

2.1 Multi-view Architecture Descriptions

Multi-view architecture descriptions have long been established as the dominant approach for structuring architectural knowledge. Standards such as ISO/IEC/IEEE 42010 define an architecture description as a collection of views, each governed by a viewpoint that addresses a specific set of stakeholder concerns [12].

This formulation has proven effective in several respects. By explicitly separating concerns, multi-view descriptions reduce local cognitive complexity, support stakeholder-specific reasoning, and enable parallel development of architectural knowledge [2, 16].

However, the standard deliberately abstracts away from certain epistemic issues. While it specifies how viewpoints are defined and how views correspond to concerns, it provides limited guidance on how architectural knowledge represented in different views is integrated, trusted, and maintained as a coherent whole. In particular, it does not address where cross-view relationships should be authoritatively represented, nor how epistemic authority is preserved as views evolve at different rates [22].

As a result, while ISO/IEC/IEEE 42010 offers a robust account of concern decomposition, it remains largely silent on how epistemic authority and integration are configured across architectural artifacts.

2.2 Limits of Coverage-Oriented Reasoning

Most architecture description theories implicitly adopt a coverage-oriented rationale: architectural quality improves as more relevant concerns are identified, more viewpoints are defined, and more artifacts are produced [12]. Under this assumption, the primary risk lies in omission.

This rationale embeds a monotonicity hypothesis: that adding viewpoints or artifacts cannot reduce the epistemic quality of an architecture description. Once issues of authority, trust, and cognitive integration are considered, however, this hypothesis no longer holds [23, 29].

In practice, increased coverage often leads to fragmentation of epistemic responsibility, duplication of relational knowledge, and ambiguity about which artifacts should be treated as authoritative [18, 34]. These effects are not incidental failures of documentation discipline; they arise systematically when responsibility for architectural knowledge is left implicit.

The limitation, therefore, is theoretical rather than empirical. Coverage-oriented reasoning lacks the conceptual machinery to explain why documentation quality may degrade as coverage increases, or why certain documentation structures remain stable while others erode under evolutionary pressure.

This gap motivates the need for a theory that addresses not only *what* architectural knowledge is documented, but *how epistemic responsibility for that knowledge—and its interconnections—is explicitly configured.*

3 Representational Responsibility as a Theoretical Primitive

3.1 Definition and Scope

To address the limitations identified in coverage-oriented reasoning, this paper introduces *Architectural Responsibility Configuration (ARCM)*, a responsibility-oriented theory that makes epistemic responsibility in architecture documentation explicit and analytically tractable.

At the core of ARCM is the concept of *representational responsibility*, introduced as a foundational theoretical primitive.

Definition 3.1 (Representational Responsibility). A representational responsibility is the epistemic obligation to authoritatively represent a set of architectural facts and the relations among them for a given audience and purpose.

This definition deliberately departs from common interpretations of responsibility in software engineering. Representational responsibility is not equivalent to artifact ownership, process role assignment, or organizational accountability, as discussed in role- or process-oriented models [13]. Rather, it concerns epistemic authority: which artifact is trusted to state what is true about the architecture, and under what conditions that trust is maintained over time.

By framing responsibility as an epistemic construct, ARCM treats architecture documentation as a system of knowledge claims rather than a collection of descriptive records. An artifact bearing representational responsibility serves as a stable reference point against which other representations can be interpreted, validated, or revised.

3.2 Responsibility for Relations, Not Only Facts

ARCM extends the notion of representational responsibility beyond individual architectural facts to encompass the relationships between them.

In multi-view architecture descriptions, many consequential architectural commitments are relational rather than atomic. Examples include deployment constraints, regulatory traceability, or the correspondence between runtime behavior and design-time assumptions. Such relational knowledge typically spans multiple viewpoints and resists encapsulation within a single view [22].

Coverage-oriented frameworks implicitly assume that these relationships will emerge from the juxtaposition of views. In practice, this assumption leads to ambiguity: relationships are redundantly described, inconsistently interpreted, or left implicit across artifacts [18].

Under ARCM, inter-view relationships are treated as first-class epistemic objects. Each such relationship must be anchored by an explicit representational responsibility—an artifact that is authoritative for stating and maintaining that relationship as the system evolves. Without such anchoring, relational knowledge becomes fragmented, contested, or silently corrupted.

3.3 Divergence of Viewpoints, Convergence of Artifacts

ARCM draws a sharp distinction between the epistemic roles of viewpoints and artifacts.

Viewpoints are inherently divergent. Each viewpoint provides a selective projection of the system, emphasizing certain concerns while abstracting away others. Such divergence is not a flaw but a necessary condition for managing complexity and supporting heterogeneous stakeholder reasoning [16].

Artifacts, by contrast, must ultimately be convergent with respect to epistemic authority. An artifact bearing representational responsibility is expected to integrate multiple distinctions into a coherent and authoritative representation. Convergence does not require collapsing all viewpoints into a single description, but it does require that responsibilities be explicitly bounded and non-overlapping.

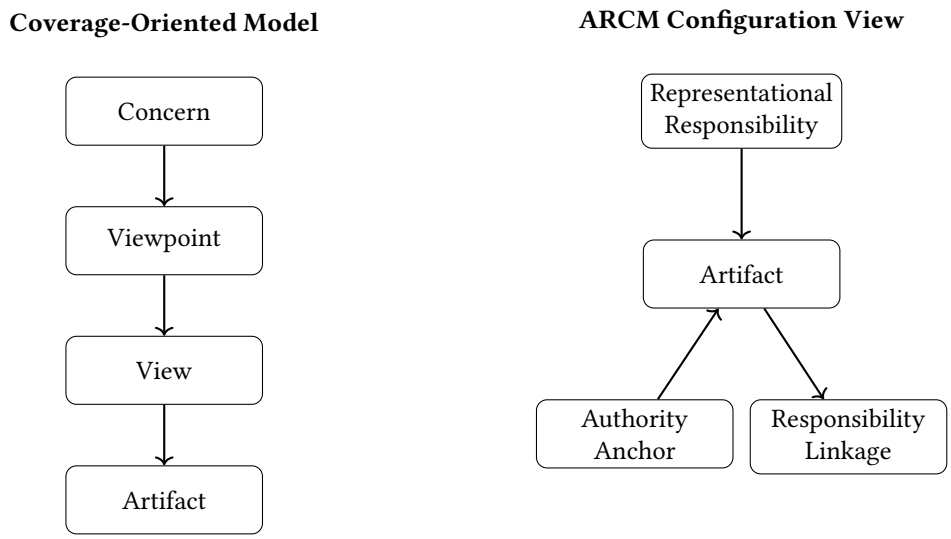

Fig. 1. Contrasting coverage-oriented architecture description with responsibility-oriented configuration.

This distinction resolves a longstanding ambiguity in architecture documentation: whether an artifact may legitimately embody multiple viewpoints. Under ARCM, an artifact may do so if and only if the representational responsibilities associated with those viewpoints are compatible and do not overlap.

Thus, ARCM permits multi-view artifacts while prohibiting duplication of epistemic authority. Overlap at the level of representation is acceptable; overlap at the level of responsibility is not. This principle provides a theoretical basis for avoiding fragmentation without enforcing artificial separation.

4 Architectural Responsibility Configuration (ARCM)

4.1 Core Constructs

Building on representational responsibility as a theoretical primitive, ARCM conceptualizes an architecture description as a *responsibility configuration* composed of four core theoretical constructs.

Artifacts. Artifacts are concrete carriers of architectural knowledge, such as documents, models, diagrams, or structured repositories. In ARCM, artifacts are not neutral containers of information; they are loci of epistemic authority, a role left largely implicit in existing architecture frameworks [12].

Responsibilities. Responsibilities denote explicit epistemic obligations assigned to artifacts. A responsibility specifies which architectural facts or relations an artifact is authoritative for representing. Responsibilities constitute the primary unit of allocation and analysis in ARCM.

Authority Anchors. An authority anchor is the artifact designated as the trusted reference point for a given responsibility. Authority anchoring determines where epistemic disputes are resolved and which representation prevails when inconsistencies arise.

Linkages. Linkages are explicit relations between responsibilities across artifacts. They capture dependencies, constraints, or correspondences without collapsing multiple responsibilities into a single authority locus.

Together, these constructs define a responsibility topology over architectural knowledge. ARCM deliberately abstracts away from notation, tooling, and organizational structure; it is concerned exclusively with how epistemic authority is configured across artifacts.

4.2 Epistemic Gravity

To explain why certain responsibility configurations tend to stabilize or collapse, ARCM introduces the concept of *epistemic gravity*.

Definition 4.1 (Epistemic Gravity). Epistemic gravity denotes the strength of semantic dependence between representational responsibilities, measured by the degree to which understanding one responsibility requires simultaneous access to another.

When responsibilities exhibit high epistemic gravity, stakeholders cannot meaningfully interpret them in isolation. Such dependencies exert convergent pressure on responsibility configuration: responsibilities with strong mutual gravity tend to require co-location within a single authority anchor or tight anchoring through explicit linkage.

Configurations that disregard epistemic gravity are structurally unstable. High-gravity responsibilities distributed across loosely connected artifacts force stakeholders to reconstruct relationships cognitively. This externalization of integration effort increases error and inconsistency [29].

Epistemic gravity is independent of organizational boundaries or tool support. It arises from the inherent structure of architectural knowledge and therefore functions as a structural force rather than a design preference.

4.3 Knowledge Metabolism Rate

While epistemic gravity explains spatial pressure on responsibility configuration, it does not account for temporal instability. ARCM therefore introduces the notion of *knowledge metabolism rate*.

Definition 4.2 (Knowledge Metabolism Rate). The knowledge metabolism rate of a representational responsibility is the frequency and magnitude with which the architectural knowledge it governs changes over time.

Responsibilities differ substantially in their metabolism. Architectural principles, regulatory constraints, and global invariants typically evolve slowly, whereas deployment details, operational parameters, or performance characteristics often change rapidly. Prior studies of software evolution have noted the destabilizing effects of combining concerns with incompatible change rhythms [3, 20].

ARCM posits that assigning responsibilities with significantly mismatched metabolism rates to a single authority anchor is structurally unstable. High-frequency change erodes trust in low-frequency knowledge, while low-frequency stabilization impedes timely evolution of volatile knowledge.

Such metabolic mismatch leads to progressive authority erosion: artifacts lose their status as trusted references not because they are incorrect, but because they cannot sustain coherence under conflicting temporal pressures.

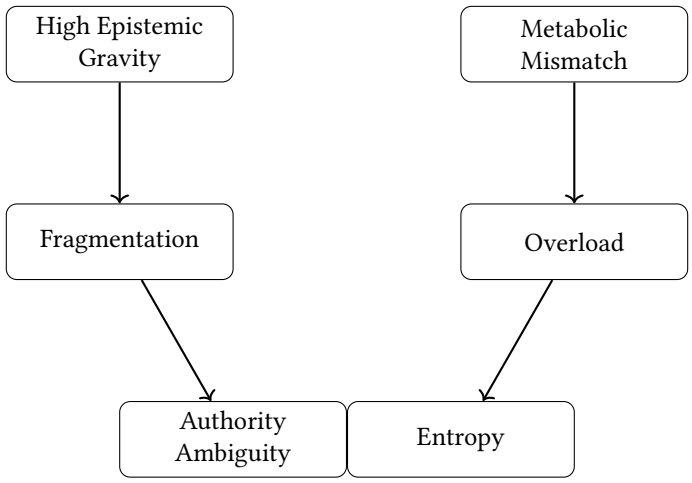

Fig. 2. Responsibility pathologies as predictable outcomes of misaligned epistemic forces.

4.4 Configuration as a Dynamic System

Epistemic gravity and knowledge metabolism rate jointly determine the dynamic stability of responsibility configurations. Architecture documentation should therefore be understood as a dynamic epistemic system subject to continuous structural forces.

Configurations that align responsibility locality with epistemic gravity and respect metabolic compatibility tend to remain stable. Configurations that violate these forces accumulate latent tension, which manifests over time as fragmentation, overload, authority ambiguity, or entropy.

ARCM does not assert that any particular configuration is universally optimal. Instead, it characterizes stability as a contingent equilibrium: each configuration embodies trade-offs between coherence, autonomy, and evolutionary flexibility. Stability is achieved not by eliminating tension, but by configuring responsibilities so that tension is structurally absorbed rather than displaced.

5 Responsibility Pathologies

5.1 Fragmentation

Fragmentation occurs when a representational responsibility exhibiting high epistemic gravity is distributed across multiple artifacts without a clear authority anchor.

Under ARCM, fragmentation is the predictable structural outcome of failing to absorb epistemic gravity within the responsibility configuration. When understanding an architectural commitment requires simultaneous access to multiple artifacts, stakeholders are forced to perform integration cognitively rather than structurally.

This cognitive reconstruction is inherently unstable. Different stakeholders assemble relationships differently, inconsistencies accumulate, and no artifact can be trusted as an authoritative reference. Fragmentation thus represents the externalization of epistemic gravity as coordination overhead rather than its structural accommodation [19, 26].

5.2 Overload

Overload arises when an artifact is assigned multiple heterogeneous responsibilities that violate metabolic compatibility, regardless of whether their epistemic gravity is low.

In such configurations, responsibilities are aggregated for convenience, historical reasons, or tooling constraints rather than epistemic alignment. The artifact becomes cognitively dense and evolutionarily fragile: changes intended for one responsibility interfere with others.

From an ARCM perspective, overload is not merely a usability concern [6, 30], but a structural failure to respect knowledge metabolism rates. Over time, overloaded artifacts resist incremental evolution and tend either to fragment or to be bypassed altogether, accelerating documentation decay.

5.3 Authority Ambiguity

Authority ambiguity occurs when no artifact is explicitly designated as the authority anchor for a responsibility, or when multiple artifacts implicitly claim such authority.

This pathology often emerges in coverage-oriented documentation strategies, where overlapping views partially represent the same architectural knowledge. In the absence of explicit responsibility anchoring, these overlaps evolve independently, producing competing representations.

Without an authority anchor, inconsistencies cannot be resolved structurally. Resolution is deferred to informal negotiation or local interpretation, leading to progressive erosion of trust in the documentation as a whole [12, 33].

5.4 Entropy

Entropy denotes the progressive loss of coherence in an architecture description over successive evolution cycles.

Unlike fragmentation or overload, entropy is not tied to a single configuration violation. It emerges when unresolved epistemic tensions—ignored gravity, metabolic mismatch, or persistent authority ambiguity—accumulate over time.

As evolution proceeds, artifacts drift semantically, linkages weaken, and responsibilities become implicit or contradictory. Entropy thus represents the temporal manifestation of unresolved structural misalignment rather than accidental degradation or individual negligence [9].

5.5 Pathologies as Structural Consequences

The responsibility pathologies identified above are not anomalies or execution failures. Under ARCM, they are predictable structural consequences of responsibility configurations that violate epistemic gravity, metabolic compatibility, or authority anchoring.

This perspective reframes documentation failure as a structural phenomenon rather than a managerial or procedural one. Improving architecture documentation therefore requires not merely additional viewpoints or stricter processes, but deliberate reconfiguration of representational responsibility.

6 Propositions: Architecture Evolution Laws

6.1 From Responsibility Configuration to Evolutionary Law

The Architecture Responsibility Configuration Model (ARCM) yields a set of explanatory propositions that characterize how architecture documentation evolves under structural epistemic constraints.

These propositions are not prescriptive rules. They are deductive laws derived from the interaction between representational responsibility, epistemic gravity, and knowledge metabolism. Each proposition specifies conditions under which a responsibility configuration is expected to remain stable, drift, or collapse under evolutionary pressure.

Crucially, each proposition is falsifiable: it can be refuted by empirical cases in which the stated structural conditions hold but the predicted outcome does not occur. This formulation follows established theory-building practice in software engineering, where explanatory adequacy and falsifiability take precedence over normative completeness [10].

6.2 P1: Responsibility Locality–Stability Law

Proposition P1 (Responsibility Locality–Stability). Architecture descriptions in which responsibilities exhibiting high epistemic gravity are locally anchored demonstrate greater long-term stability under evolution.

Explanation. Responsibility locality holds when a responsibility and the relations it depends on are anchored within the same authoritative artifact or tightly coupled authority cluster. Under this condition, changes can be absorbed locally without repeated cross-artifact reconciliation.

When locality is violated, epistemic gravity is externalized as coordination overhead, leading to fragmentation and authority ambiguity. This proposition refines classic locality arguments in modular design by making epistemic authority explicit [1, 24].

Falsifiability. This proposition is falsified if documentation systems consistently maintain coherence and low coordination cost despite distributing high-gravity responsibilities across loosely coupled artifacts.

6.3 P2: Metabolic Compatibility Law

Proposition P2 (Metabolic Compatibility Law). Co-locating responsibilities with significantly mismatched knowledge metabolism rates leads to progressive authority erosion.

Explanation. Responsibilities evolving at different temporal rhythms impose incompatible maintenance pressures when aggregated. High-frequency updates obscure slow-changing normative knowledge, while rigid constraints inhibit timely evolution of volatile knowledge.

Over time, such artifacts lose perceived authority—not because they are incorrect, but because they cannot maintain coherence under conflicting temporal demands.

Falsifiability. This proposition is refuted if an artifact persistently maintains authority while co-locating responsibilities with strongly mismatched metabolism rates.

6.4 P3: Relational Responsibility Anchoring Law

Proposition P3 (Relational Responsibility Anchoring). Responsibilities for inter-view relations must be anchored in artifacts that co-evolve with the views they relate.

Explanation. Architectural knowledge is constituted not only by isolated facts but by relations across viewpoints. When relational responsibilities are anchored in evolutionarily decoupled artifacts, inconsistencies accumulate and authority ambiguity emerges.

Prior work on multi-view consistency has addressed this problem as a tooling or synchronization issue; ARCM reframes it as a responsibility allocation problem [5].

Falsifiability. This proposition is falsified if cross-view relations remain coherent over time despite being anchored in artifacts that do not co-evolve with the related views.

6.5 P4: Epistemic Complexity Conservation Law

Proposition P4 (Epistemic Complexity Conservation). The total epistemic complexity of an architecture description is conserved and redistributed between internal aggregation cost and external linkage cost.

Explanation. Unified configurations internalize complexity through aggregation, increasing internal coupling while reducing navigational overhead. Federated configurations externalize complexity through explicit linkages, preserving autonomy at the cost of navigation and coordination.

No responsibility configuration eliminates epistemic complexity; it only redistributes where and how that complexity is paid. This reframes classical essential–accidental complexity distinctions as a conservation law over responsibility configurations [4].

Falsifiability. This proposition is contradicted if a responsibility configuration simultaneously minimizes both internal aggregation cost and external linkage cost without incurring trade-offs elsewhere.

6.6 Propositions as Generative Constraints

Each responsibility pathology identified in Section 5 corresponds to a violation of one or more propositions: fragmentation violates P1, overload violates P2, authority ambiguity violates P3, and entropy emerges from unresolved cumulative violations.

Together, these propositions define a closed explanatory system. Given a responsibility configuration, ARCM predicts the dominant evolutionary pressures and the equilibrium states toward which the system tends to converge.

7 Responsibility Patterns as Epistemic Equilibria

The patterns identified by ARCM are not design prescriptions. They are stable epistemic equilibria: responsibility configurations toward which architecture documentation systems tend to converge under sustained evolutionary pressure.

Unlike conventional architecture patterns, which encode reusable structural solutions, ARCM patterns describe equilibrium responses to epistemic gravity, metabolic constraints, and authority distribution. Each pattern represents a locally stable resolution of the laws articulated in Section 6.

7.1 Pattern A: Unified Authority Core

Structure. A small set of central artifacts concentrates representational responsibility for multiple viewpoints and their relations.

Epistemic Equilibrium. This configuration maximizes responsibility locality (P1). High-gravity responsibilities are co-located, minimizing coordination overhead and ambiguity.

Stability Conditions. The pattern is stable when:

- The number of viewpoints is limited;
- Knowledge metabolism rates are aligned (P2);
- Change is infrequent or centrally governed.

Failure Mode. As scope expands or metabolism diverges, overload and authority erosion emerge due to violation of metabolic compatibility (P2).

7.2 Pattern B: Federated Responsibility Mesh

Structure. Responsibilities are distributed across multiple artifacts with localized authority. Interview relations are externalized as explicit, owned linking responsibilities.

Epistemic Equilibrium. This pattern minimizes aggregation cost while externalizing complexity through navigable relations, consistent with the Complexity Conservation Law (P4).

Stability Conditions. The pattern is stable when:

- Viewpoints evolve at different rates;
- Authority boundaries are respected;
- Relational responsibilities are explicitly anchored and maintained (P3).

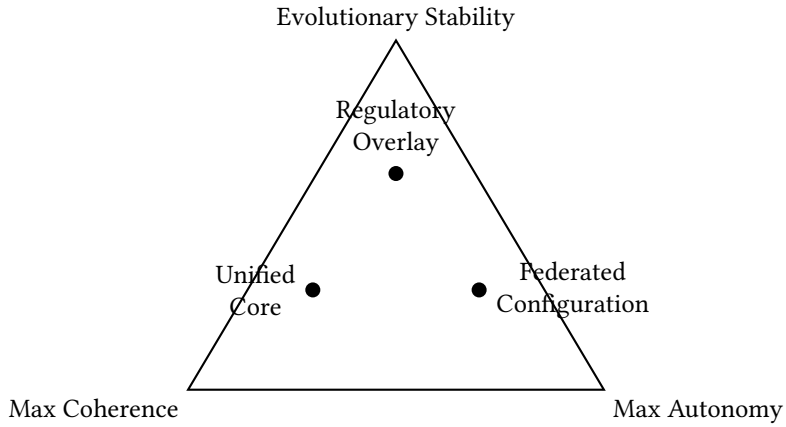

Fig. 3. Epistemic configuration space illustrating stable responsibility topologies and conserved tensions.

Failure Mode. Fragmentation and authority ambiguity arise if relational responsibilities are implicit or evolutionarily decoupled (violation of P3).

7.3 Pattern C: Regulatory Responsibility Overlay

Structure. A stable overlay of low-metabolism, normative responsibilities constrains multiple high-velocity operational artifacts without absorbing their change pressure.

Epistemic Equilibrium. This pattern isolates metabolic incompatibility by stratifying authority rather than aggregating responsibilities, satisfying P2 without violating P1.

Stability Conditions. The pattern is stable when:

- Overlay responsibilities change slowly;
- Authority boundaries are explicit;
- Cross-layer relations are unidirectional and minimal.

Failure Mode. Entropy emerges when overlay constraints drift into operational artifacts or when operational volatility contaminates normative authority.

7.4 Pattern Trade-offs and Conservation

Each pattern represents a distinct equilibrium point in the conserved epistemic trade space defined by Proposition P4. Unified cores internalize complexity, federated meshes externalize it, and regulatory overlays stratify it temporally.

No pattern dominates across all dimensions. Stability arises not from optimality, but from alignment between responsibility configuration and epistemic forces. Earlier architectural trade-off analyses have noted the persistence of competing forces among coherence, autonomy, and evolvability, typically framed as design decision spaces; ARCM reinterprets these observations as manifestations of conserved epistemic complexity rather than discretionary trade-offs [15].

7.5 Patterns as Predictive Constructs

ARCM patterns do not prescribe architectural best practices. They predict how responsibility configurations tend to stabilize or degrade under evolution.

Practitioners may intentionally choose configurations aligned with desired trade-offs, but ARCM makes explicit the structural costs and failure modes such choices entail.

8 Retrospective Illustrative Application

This section employs abductive reasoning to reconstruct the most plausible responsibility configurations consistent with observed architectural artifacts. Abduction is used not for empirical validation, but for explanatory adequacy: assessing whether ARCM can coherently account for documented failures and evolutionary trajectories that coverage-oriented theories leave unexplained.

8.1 Methodological Positioning

This study adopts a retrospective, abductive analysis of published and practitioner-reported architecture documentation cases. The purpose is not to test propositions statistically, but to evaluate whether ARCM provides a parsimonious and structurally grounded explanation of phenomena that recur across diverse contexts.

Cases are treated as epistemic traces rather than datasets. The analysis asks whether ARCM can reconstruct latent responsibility configurations that plausibly generated the observed documentation outcomes. This abductive use of cases follows established practices in theory-oriented software engineering research, where cases probe explanatory scope rather than empirical generality [27, 31].

8.2 Case Selection Rationale

Cases were selected to satisfy three criteria:

- Presence of explicit multi-view architecture documentation;
- Reported difficulties in comprehension, coordination, or evolution despite high coverage;
- Sufficient descriptive detail to reconstruct artifact responsibilities and authority relations.

The selected cases span large-scale enterprise systems, safety-critical platforms, and long-lived software ecosystems, ensuring variation in organizational structure, governance style, and evolutionary pressure.

8.3 Abductive Reconstruction of Responsibility Configurations

For each case, architectural artifacts were reinterpreted through the ARCM lens via abductive reconstruction. Rather than treating documented views and artifacts at face value, the analysis inferred the implicit responsibility configuration governing architectural knowledge.

This form of reinterpretation aligns with prior work that re-analyzes architecture documentation to surface latent knowledge structures and rationale not explicitly represented [14].

Specifically, the reconstruction focused on identifying:

(1) Representational responsibilities claimed or implicitly assumed by each artifact;
(2) Authority anchors governing both factual and relational architectural knowledge;
(3) Explicit or implicit linkages between viewpoints and their responsibility ownership.

This process frequently revealed responsibility configurations that were not explicitly documented, yet nonetheless shaped architectural reasoning, conflict resolution, and long-term evolution.

8.4 Observed Pathologies as Structural Outcomes

Across cases, ARCM pathologies manifested consistently as structural outcomes rather than individual failures.

Fragmentation emerged when high-gravity relational responsibilities were distributed across artifacts without explicit authority anchoring. Overload occurred when artifacts accumulated heterogeneous responsibilities with incompatible knowledge metabolism rates. Authority ambiguity was prevalent where multiple artifacts implicitly claimed interpretive control over the same relations. Entropy appeared as a gradual loss of trust in documentation as an epistemic reference.

These phenomena were not attributable to insufficient viewpoints or missing concerns, but to misaligned responsibility configurations. Comparable degradation patterns have been reported across domains, though typically attributed to process failure or tooling limitations rather than structural responsibility misalignment [9, 21].

8.5 Evolutionary Trajectories Across Configurations

Several cases exhibited consistent evolutionary trajectories across epistemic topologies. Early-stage systems tended toward Unified Authority Cores, favoring coherence and centralized authority. As scale and change pressure increased, configurations shifted toward Layered Stratification or Federated Meshes.

These transitions were reactive rather than intentional, driven by rising metabolic mismatch and authority overload. In the absence of explicit responsibility reconfiguration, transitions often passed through unstable hybrid states with elevated entropy. Similar staged transitions have been observed in longitudinal studies of large-scale systems, though without a structural explanation grounded in responsibility configuration [20].

Taken together, these cases do not constitute empirical proof of ARCM's propositions. Instead, they serve as stress tests for explanatory adequacy, motivating a comparative discussion of what ARCM explains that existing architecture description theories cannot.

9 What ARCM Explains That Others Cannot

Taken together, the preceding analysis positions ARCM at a distinct theoretical tier. Where ISO/IEC/IEEE 42010 provides a normative classification framework for architecture description, and ATAM characterizes architectural decision spaces for evaluative purposes, ARCM operates as an explanatory theory: it introduces irreducible epistemic constructs and derives falsifiable propositions that account for the stability and decay of architecture documentation under evolution.

9.1 Limits of Coverage-Oriented Architecture Theories

Coverage-oriented architecture theories, including viewpoint-based frameworks and standards such as ISO/IEC/IEEE 42010, explain documentation quality primarily in terms of concern identification and representation completeness [12]. Within this paradigm, documentation failure is attributed to missing views, inadequate stakeholder analysis, or insufficient update discipline [11].

However, these explanations break down in cases where coverage is demonstrably high yet documentation remains fragmented, distrusted, or unusable. Empirical and experience-based studies repeatedly report that adding views and artifacts can increase confusion rather than reduce it [9, 21]. Coverage-oriented theories lack constructs for reasoning about epistemic authority, inter-view relations, and responsibility allocation.

ARCM explains these failures by identifying responsibility misconfiguration as a causal dimension orthogonal to coverage.

9.2 Why Process- and Governance-Centered Accounts Fall Short

Process-oriented accounts explain documentation decay through breakdowns in compliance, review practices, or organizational discipline [17]. Governance models emphasize ownership, role assignment, and lifecycle management [32].

While such accounts capture important contributing factors, they implicitly assume that correct processes can compensate for any structural arrangement of artifacts. ARCM challenges this assumption by showing that no amount of process rigor can stabilize configurations that violate epistemic locality, metabolic compatibility, or authority anchoring.

In ARCM, process quality modulates outcomes but does not determine them; responsibility configuration constrains what processes can achieve.

9.3 Explaining Authority, Trust, and Relational Knowledge

Existing theories offer limited tools for explaining why stakeholders distrust architecture documentation even when it is factually accurate. Studies of architectural knowledge management report persistent problems of trust, interpretation conflict, and unclear authority [14].

ARCM accounts for this phenomenon by distinguishing factual correctness from epistemic authority. Trust depends on stable authority anchors for both facts and relations. When relational knowledge lacks a clear responsibility owner, stakeholders cannot determine which artifacts to trust, regardless of internal consistency.

This explanatory capability is absent from coverage- and process-oriented models.

9.4 Explaining Evolutionary Decay Without Human Blame

Documentation decay is often attributed to human factors such as negligence, time pressure, or organizational churn [4]. ARCM offers an alternative explanation grounded in structural inevitability.

As systems evolve, epistemic gravity and knowledge metabolism exert continuous pressure on static configurations. Without explicit responsibility reconfiguration, even well-maintained documentation accumulates fragmentation, overload, or authority ambiguity.

ARCM thus explains decay as a predictable outcome of unresolved epistemic forces rather than individual or organizational failure.

9.5 Structural Necessity Versus Best Practices

Many architecture frameworks implicitly promote best practices, suggesting that correct application leads to success [15]. ARCM rejects this normative stance.

By modeling architecture documentation as an epistemic system governed by conservation of complexity and responsibility allocation, ARCM shows that all configurations involve trade-offs. Stability is achieved not by universal practices, but by alignment between responsibility configuration and epistemic forces.

9.6 Summary of Explanatory Advantages

In summary, ARCM explains:

- Why high viewpoint coverage can reduce coherence;
- Why authority ambiguity undermines trust independently of accuracy;
- Why documentation decay persists despite process improvements;
- Why stable configurations emerge as responses to epistemic tensions rather than design choices.

These explanatory capabilities are not derivable from existing architecture description theories, indicating that responsibility configuration constitutes a missing theoretical dimension.

10 Threats to Validity

10.1 Construct Validity

The central construct introduced in this paper is *representational responsibility*. A potential threat concerns whether this construct is sufficiently distinct from existing notions such as ownership, accountability, or process roles [32].

ARCM explicitly defines responsibility as an epistemic construct: authoritative representation of architectural facts and relations. It does not denote organizational control or procedural obligation.

This distinction is enforced through formal definition and explanatory use, constraining interpretive drift.

10.2 Explanatory Versus Empirical Validity

ARCM is a theory-building contribution rather than an empirical generalization [10]. Its propositions are intended as falsifiable explanations of observed phenomena, not statistical claims about prevalence or effect size.

A threat arises if illustrative cases are misinterpreted as empirical validation. To mitigate this, cases are used abductively to demonstrate explanatory reconstruction rather than hypothesis testing [31]. Empirical validation is explicitly deferred as future work.

10.3 Scope and Boundary Conditions

ARCM focuses on architecture documentation as an epistemic system embodied in artifacts. It does not model runtime system behavior, informal communication, or individual cognition.

Failures attributable solely to human error, organizational politics, or tooling defects therefore fall outside its explanatory scope. This boundary is intentional and reflects the theory's focus on structural configuration rather than social causation.

10.4 Generalizability of Configuration Patterns

The configuration patterns identified in this paper are not claimed to be exhaustive or universally optimal. They represent stable responses to specific epistemic tensions under identifiable conditions.

A threat arises if these patterns are interpreted as prescriptive best practices. ARCM instead frames them as equilibrium points in a configuration space governed by conservation laws.

10.5 Summary

The primary threats to validity concern construct interpretation and scope rather than empirical uncertainty. By clearly delimiting responsibility as an epistemic construct and positioning ARCM as an explanatory theory, the paper mitigates these threats while preserving falsifiability.

11 Conclusion

11.1 Theoretical Closure

This paper addressed a persistent phenomenon: architecture documentation frequently fails to sustain comprehension, coordination, and authority over time, even when it achieves comprehensive viewpoint coverage. Existing architecture description frameworks lack the theoretical constructs required to explain this failure.

By introducing the Architecture Responsibility Configuration Model (ARCM), this work reframes architecture documentation as a dynamic epistemic system governed by responsibility topology, epistemic gravity, knowledge metabolism, and authority anchoring.

This reframing closes a long-standing theoretical gap by elevating responsibility configuration from an implicit concern to an explicit explanatory construct, enabling law-like reasoning about documentation stability and decay.

11.2 Falsifiability and Explanatory Power

ARCM is not proposed as a normative framework or a set of best practices, but as an explanatory theory with falsifiable propositions [25]. Its claims can be refuted by demonstrating architecture descriptions that violate the proposed configuration laws while remaining epistemically stable over time.

Conversely, ARCM demonstrates explanatory power by accounting for failure modes that are systematically invisible to coverage-oriented and process-centric theories.

11.3 Future Research Space

ARCM opens multiple directions for future research. Empirical studies may operationalize responsibility configurations to test propositions across domains. Tool support may explore how responsibility anchoring and linkage visibility can be made explicit in documentation environments.

Beyond software architecture, the model invites investigation into other large-scale knowledge systems where authority, evolution, and representation interact.

More broadly, this work suggests that the long-term viability of architectural knowledge depends not on the accumulation of views, but on the explicit and evolving configuration of responsibility within epistemic systems.

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